

Mega-Cap Dominance and Systematic Risk: Implications for Beta and Implied Volatility

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Abstract

This paper examines the impact of market concentration on systematic risk, focusing on the S&P 500's "Magnificent Seven" (Mag7). Using a weighted least squares regression with exponential decay on weekly returns, we find that the Mag7 contribute over 40% of the index beta in short-term horizons, reducing diversification benefits and amplifying downside risk. The dominance of these stocks distorts option pricing, as concentration effects steepen volatility skews and elevate tail risk in index derivatives. Further, our empirical analysis reveals that traditional long small-cap/short large-cap strategies underperform in current market dynamics, while a beta-neutral approach incorporating high-growth mega caps enhances risk-adjusted returns. These findings highlight the need for investors to reconsider the definitions and measurements of "market risk".

1 Introduction

The increasing concentration of the S&P 500 has fundamentally reshaped its risk profile, with a handful of mega-cap technology stocks—the "Magnificent Seven" Mag7—now dominating index performance. This shift has amplified systematic risk, reducing diversification benefits and distorting traditional market risk measures. Normalized entropy—a measure of weight distribution—has fallen from a 15-year average of 0.8832 (2005–2020) to 0.8099 as of 2025 (Figure 10), indicating reduced diversification and a growing imbalance in index composition. This concentration heightens systemic risk and questions the S&P500's reliability as a diversified benchmark. This paper examines the implications of this concentration through multiple dimensions. Section (2) analyzes the role of Mag7 stocks in driving S&P 500 beta, evaluating their contribution to systematic risk and their impact on return forecasting. Section (3) explores how this concentration affects option pricing and implied volatility, using empirical analysis and synthetic basket modeling to assess market distortions. Section (4) investigates the challenges this environment poses to traditional long small-cap/short large-cap investment strategies and proposes a beta-neutral alternative that enhances risk-adjusted returns. Finally, Section (5) synthesizes our findings, discussing broader implications for investors and avenues for future research.

2 Analysis of Beta in the S&P 500

Market beta measures an asset's sensitivity to a benchmark—typically the cap-weighted S&P 500—and quantifies its systematic (non-diversifiable) risk versus idiosyncratic risk. It is commonly calculated as:

$$\beta_i = \frac{\text{Cov}(r_i, r_m)}{\text{Var}(r_m)} = \frac{\sigma_i}{\sigma_m} \rho_{i,m}$$

This can be determined using standard deviations and the correlation between asset and market returns, or via least squares regression. In our research, we modify this approach to yield a more accurate market beta estimation. This section explores the implications of the concentration of the Magnificent 7 (Mag 7) in the S&P 500, with a focus on beta. In particular, we address the following three subsections:

1. Three different beta estimates
2. The effect on forecasting returns
3. Empirical observations of the TSLA drop

2.1 Beta Estimation

A. Computation of Beta

To analyze the beta of the Mag 7 collectively, we construct a cap-weighted portfolio using free-floating market capitalizations as of January 31, 2025, while keeping the weights constant for our beta evolution analysis.

B. Data

We obtained daily and weekly price data for the Mag 7 from Yahoo Finance and computed total logarithmic returns (including dividends). Log returns were chosen due to their near-normal distribution, reduced skewness, and stabilized variance. The Ljung-Box test[1] confirmed significant autocorrelation in daily data—likely due to market microstructure noise—whereas weekly data provided more independent observations and improved model reliability while still capturing recent market trends.

C. Framework

We compute the market beta for the Mag 7 portfolio using a Weighted Least Squares (WLS) regression rather than classical Ordinary Least Squares[2]:

$$\min_{\beta} \sum_{i=1}^n w_i (\alpha - X_i \beta)^2$$

The rationale includes taking into account the nature of our data. It is known that when handling financial time series data, more recent observations are more relevant than older observations[3]. Weighted least-squares regression allows us to apply an exponential time decay to our observations, gradually reducing their influence as they age. Recognizing that recent data more accurately represents current risk exposures, we apply exponentially decaying weights, $w_i = \lambda^{(T-i)}$ to past observations. For short-term regression, we use $\lambda = 0.9967$, assigning 50% weight to the oldest observation, while for long-term regression, we use $\lambda = 0.9832$, giving roughly 35% weight. These decay parameters balance the need to emphasize recent market dynamics—crucial for near-term risk assessment—and to retain older observations for capturing long-run trends. The quicker decay for weekly short-term regression reflects the higher volatility and rapid changes in short-horizon returns, whereas the lower decay in long-term regression aligns with a longer-term perspective.

Table 1: Regression Results for the Mag 7 Portfolio

Freq.	Method	Alpha	Beta	R ²	Res. Std.	Beta ~493	Contrib. (%)
5Y Monthly	Cov/Var	—	1.2680	—	—	—	—
	WLS	0.0098	1.2752	0.635	0.0486	0.886	36.51
18M Weekly	Cov/Var	—	1.4998	—	—	—	—
	WLS	0.0012	1.4237	0.696	0.0164	0.832	40.50

D. Results

Table 1 reports the Mag 7 portfolio’s beta estimates relative to the S&P500 over two horizons. For 5-year regression, both the simple and WLS methods yield betas around 1.27-1.28, indicating moderate market sensitivity. Over the shorter term, betas rise to 1.42-1.50, suggesting that these stocks have recently become more influential in driving the overall market. The effect of concentration is evident in the contribution to market beta: over the long-term the Magnificent Seven accounted for 36.51% of the total index beta and over 40% in the shorter time frame. This growing influence reduces diversification benefits and implies that systematic risk is increasingly determined by a small subset of mega-cap firms.

2.2 Effect on Forecasting Returns

Understanding that the Magnificent Seven are the primary drivers of market beta, we examine how the Magnificent Seven’s outsized influence on market beta affects return forecasts and systemic risk. To do this, we build a basket of Mag7 plus energy and real estate stocks, sectors identified through a historical correlation analysis of SPDR ETFs as least correlated with technology. By contrasting sectors with varying tech exposure, we illustrate how investment evaluations shift under heavily concentrated market conditions. We use weekly price data to calculate logarithmic returns for each stock, as well as for the cap-weighted S&P500 and the Invesco Equal-Weighted S&P500. We then compute betas against both benchmarks using a WLS (exponential decay) method outlined in Part 1. Our findings show that Mag 7 stocks have significantly higher betas relative to SPX, reflecting the large weights in the cap-weighted index. In contrast, energy and real estate stocks tend to have higher betas relative to RSP, where equal weighting reduces mega-cap technology’s dominance. Next, we calibrate the first two moments (drift and volatility) of SPX and RSP from ten years of historical weekly log returns, then use these parameters to simulate future price paths via Geometric Brownian Motion[4]:

$$S_{t+\Delta t} = S_t \exp \left[\left(\mu - \frac{\sigma^2}{2} \right) \Delta t + \sigma \epsilon \sqrt{\Delta t} \right].$$

We assume an initial price equal to the last trading price of 2024 for each index and generate 50,000

simulated paths to obtain expected returns calculated by averaging the terminal values of the simulated paths.

A. CAPM

To isolate the effect of varying beta calculations on return forecasts, we average the SPX and RSP expected returns to obtain $E[R_m] = 11.13$. We then apply the CAPM formula[5],

$$\mu_i = r_f + \beta_i(E[R_m] - r_f),$$

This approach highlights the discrepancies that arise when using a cap-weighted versus an equal-weighted index as the benchmark. We obtain the 1-Year Constant Maturity Rate from the Federal Reserve Economic Data[6] as the risk-free rate. Based on our findings, technology stocks show higher expected returns when their betas are measured against SPX, whereas non-tech stocks appear relatively less risky ($\beta < 1$) because Mag 7 stocks dominate systemic risk. However, a lower beta in a tech-concentrated market does not ensure lower total risk—sector-specific shocks can still drive volatility. For example, a financial stock with $\beta \approx 0.8$ may understate its exposure to industry-level events. Table 2 summarizes each stock’s beta and expected return estimates relative to SPX and RSP. Notably, Mag7 stocks (e.g., NVDA, TSLA) exhibit higher betas and expected returns against the cap-weighted index, while energy and real estate names (e.g., XOM, CVX, WELL) record higher values when measured against the equal-weighted benchmark. These patterns emphasize how market concentration can distort risk metrics and highlight the need to account for benchmark composition when forecasting returns and assessing systematic risk.

Table 2: Beta Estimates and Expected Returns Relative to SPX and RSP

Ticker	Beta (SPX)	Beta (RSP)%	Exp. Ret. (SPX)	Exp. Ret. (RSP)%
AAPL	1.178	0.776	12.36	9.57
MSFT	1.111	0.705	11.90	9.08
NVDA	1.972	1.223	17.88	12.68
GOOG	1.157	0.775	12.22	9.56
META	1.455	0.952	14.29	10.79
AMZN	1.365	0.912	13.67	10.52
TSLA	2.106	1.468	18.82	14.39
XOM	0.636	0.883	8.72	10.32
CVX	0.763	0.943	9.48	10.74
COP	0.903	1.099	10.45	11.82
ET	1.087	1.204	11.74	12.55
PLD	0.994	0.966	11.09	10.89
AMT	0.714	0.763	9.14	9.48
WELL	0.972	1.119	10.94	11.95
SPG	1.481	1.641	14.47	15.58

B. GBM Simulation of Stocks

We use a GBM framework for stock-return simulations, deriving drift from CAPM-based expected returns and volatility from historical data. The simulated returns align closely with CAPM, but SPX-based simulations show noticeably fatter tails, reflected in a wider 1st–99th quantile spread and higher excess kurtosis than their RSP-based counterparts. This heightened tail risk under SPX (Figure 9) emphasizes the importance of benchmark selection in risk management, stress testing, and hedging. Overall, our results underscore how the choice of benchmark in beta estimation can critically affect both risk and return forecasts.

2.3 Empirical Observation of TSLA Drops in 2024

Figure 1 highlights two distinct periods of Tesla’s decline in 2024:

- First drop (late December 2023 – April 2024)
- Second drop (July – August 2024)

During these drops, the other M7 stocks behave differently, with the first drop showing divergence among them, while the second drop exhibits more synchronized movement.

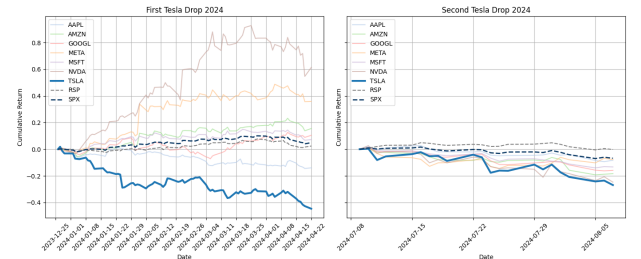


Figure 1: Tesla drops in 2024

A. Tesla’s Idiosyncratic Risk

The first Tesla decline shows a prolonged downtrend with a cumulative loss exceeding 40%, while most other M7 stocks continued to rise. This divergence underscores Tesla’s heightened idiosyncratic risk, which was driven by company-specific factors rather than broader market conditions. Notably, SPX and the equal-weighted S&P 500 (RSP) remained stable, reinforcing that this decline had minimal impact on overall market sentiment.

B. SPX Performance and Systematic Risk

The second Tesla drop coincides with a more synchronized decline across the M7 stocks, reflecting

the increasing correlation within this highly concentrated group. Unlike the first drop, where Tesla's decline was largely isolated, this event demonstrates how the collective movement of M7 stocks contributed to a broader market downturn. Given their outsized weighting in SPX, their simultaneous decline exerted a significant impact on the index, amplifying systematic risk

This synchronized movement is not simply a spillover from Tesla's decline but rather a reflection of the interconnected nature of M7 stocks, particularly their co-investment in AI and high-growth technology sectors. As these firms share similar market drivers and exposure to macroeconomic conditions, their simultaneous downturn reinforces the risk that high concentration and strong intra-group correlations pose to overall market stability.

C. RSP vs. SPX: The Role of Concentration Risk

A key distinction between the two periods is the difference in performance between SPX and RSP. While RSP remained relatively resilient during the second Tesla drop, SPX exhibited a sharper decline, underscoring the disproportionate impact of market-cap concentration. This divergence highlights that when all M7 stocks decline simultaneously, SPX is more vulnerable due to its heavy reliance on these few stocks, whereas RSP, being equal-weighted, mitigates the impact of any single group of stocks.

This highlights the risk of SPX's heavy reliance on highly correlated M7 stocks, where their collective decline amplifies market-wide volatility. In contrast, RSP's diversified structure offers greater stability, mitigating the impact of M7 downturns.

D. Implications for Market Stability

Tesla's 2024 declines illustrate how high market concentration and rising M7 correlations amplify systematic risk. While its first drop was isolated, the second coincided with broader market stress, highlighting the need to monitor correlation shifts and concentration risks.

Despite Tesla's high beta (Table 2), its declines did not always spread to other M7 stocks, showing that systematic risk coexists with firm-specific shocks. Investors must manage both market-wide volatility and idiosyncratic risks, even among the most influential index constituents.

3 Impact of Index Concentration on Option Pricing and Implied Volatility

Large-cap tech stocks, particularly the Magnificent 7, contribute significantly to the S&P 500's beta, amplifying systematic risk and leading to higher implied volatility, particularly in the downside skew. As options are fundamentally instruments for trading volatility, this concentration raises important questions that will be examined in this section:

1. What are the implications for option pricing and the volatility of the option when the index is highly concentrated?
2. How does a significant increase in the volatility of one or several underlying securities affect the option and its implied volatility?

We address these questions by constructing synthetic baskets to analyze the impact of index concentration on implied volatility and conducting an empirical study to examine historical volatility patterns and market responses.

3.1 Basket Analysis

A. Data and Preprocessing

The Russell 2000 is more diversified than the S&P 500 as evidenced by its lower market concentration - its 10 largest companies contribute less than 5.4% of its total weight, compared to roughly 31% from the Mag 7 in the S&P 500[7]. Given this diversification, we obtain option quotes for Mag 7 stocks and index options from both indices using Bloomberg data[8].

Constituent weights are approximated using market capitalization data from Bloomberg and adjusted to reflect the rebalancing schedule of the S&P 500 [9], acknowledging that the index uses free-floating market cap weights. In addition, a historical correlation matrix, ρ_{ij} , is computed from weekly log returns to capture the co-movements among the stocks, and the risk-free rate is sourced from FRED's 3-month Treasury yield (TB3MS)[10].

B. Methodology Framework

We evaluate the impact of index concentration on option pricing by integrating single-name volatility surfaces, basket aggregation, and correlation analysis. Our objective is to assess how the dominance few high-volatility securities (the Mag 7) influence the overall implied volatility. We adopt

a dual approach: (i) constructing the basket's implied volatility from its constituents, and (ii) decomposing the index volatility into contributions from the concentrated (Mag 7) and residual stocks.

i) Calibration of Volatility Surfaces We calibrate both the *implied* and *local* volatility surfaces under a risk-neutral framework. The *implied volatility surface*, obtained by inverting the Black-Scholes formula, captures market consensus on option risk premia, while the *local volatility surface*, derived via Dupire's formula [11], reflects instantaneous, state-dependent dynamics. This dual calibration combines market-driven pricing with the path-dependent behavior inherent in advanced derivative models.

- **Implied Volatility Surface:** For European call options, the market-implied volatility $\sigma_{\text{imp}}(K, T)$ is extracted by inverting the Black-Scholes formula and numerically solving

$$C_{\text{market}}(K, T) = C(S_0, K, T, r, \sigma_{\text{imp}}(K, T)) \quad (1)$$

While commonly used for pricing standard options, implied volatility does not directly reflect the volatility evolution with the underlying price.

- **Local Volatility Surface:** To capture the instantaneous, state-dependent volatility, we employ Dupire's local volatility framework [11]:

$$\sigma_{\text{local}}^2(K, T) = \frac{2 \frac{\partial C}{\partial T}(K, T)}{K^2 \frac{\partial^2 C}{\partial K^2}(K, T)}. \quad (2)$$

To mitigate data sparsity and noise, Gaussian smoothing and interpolation techniques [12] are employed during preprocessing. This local volatility surface can exhibit spikes or troughs, highlighting regions where the market implicitly prices in particularly high or low volatility for a given strike-maturity combination.

ii) Basket Volatility Construction To isolate concentration effects, the S&P 500 is decomposed into two baskets: one comprising the Mag 7 and a residual basket. We estimate the Mag 7 basket's implied volatility using two cross-validated methods:

- 1. Monte Carlo Simulation:** Each constituent's price is simulated under a correlated Geometric Brownian Motion (GBM) with state-dependent local volatility. For stock i , the evolu-

tion is modeled as

$$S_i(t + \Delta t) = S_i(t) \exp\left[\left(r - \frac{1}{2}\sigma_{\text{local},i}^2\right)\Delta t + \sigma_{\text{local},i}\sqrt{\Delta t} Z_i\right]$$

Z_i are standard normal variables generated such that their joint distribution reflects the correlation matrix ρ_{ij} via Cholesky decomposition. The basket price at maturity T is given by

$$S_{\text{basket}}(T) = \sum_{i=1}^n w_i S_i(T)$$

European call options on the basket are then priced, and the Black-Scholes inversion yields the synthetic volatility curve.

- 2. Analytical Aggregation:** Alternatively, the basket's implied volatility is computed point-wise as

$$\sigma_{\text{basket}}(K, T) = \sqrt{\sum_{i=1}^n \sum_{j=1}^n w_i w_j \sigma_i(K, T) \sigma_j(K, T) \rho_{ij}}$$

where w_i are constant weights (over a 3-month period), $\sigma_i(K, T)$ are the individual implied volatilities, and ρ_{ij} are the historical correlations.

- iii) Decomposition of Index Volatility** To isolate the contribution of the Mag 7 stocks from the rest of the S&P 500, we model the total S&P 500 implied volatility as a weighted combination of the Mag 7 basket volatility (σ_{basket}) and the Residual basket volatility (σ_{rest}). Denoting the aggregate weight of the Mag 7 by w_{M7} , S&P 500's total variance is approximated as

$$\begin{aligned} \sigma_{\text{SPX}}^2 \approx & \left(w_{\text{M7}} \sigma_{\text{basket}}\right)^2 + \left((1 - w_{\text{M7}}) \sigma_{\text{rest}}\right)^2 \\ & + 2 w_{\text{M7}} (1 - w_{\text{M7}}) \rho \sigma_{\text{basket}} \sigma_{\text{rest}} \end{aligned}$$

where σ_{rest} is the residual basket's implied volatility and ρ is the correlation between the two baskets. We solve for σ_{rest} numerically using Brent's method [13] to ensure robust estimation, even when the quadratic discriminant is small. Although our approach uses historical correlations to capture interdependence, a more refined method could derive basket-implied correlations [14]. For our analysis, the historical approximation provides an effective framework.

This framework allows us to back out the implied volatility of σ_{rest} using a numerical approach (via Brent's method[13]), ensuring a robust estimation even when the quadratic discriminant is small.

In our framework, the residual basket’s implied volatility is inferred from the observed S&P 500 volatility and the simulated Mag 7 volatility. We approximate the interdependence using historical correlations. While a basket-implied correlation could enhance accuracy [14], the historical approach offers a simplified means to assess the concentration effect.

C. Empirical Findings: Volatility Surfaces and Basket Decomposition

This section presents key findings on index concentration and volatility. We first compare the local and implied volatility surfaces of the S&P 500 and the Russell 2000, then decompose the S&P 500 into two baskets to assess how a small group of high-volatility stocks influences overall index risk.

i) SPX vs. RUT – Concentration Effects, Systematic and Idiosyncratic Risk: Figure 2 displays the local and implied volatility surfaces for the S&P 500 and the Russell 2000. The S&P 500, driven by the Mag 7, shows sharper spikes and a steeper volatility skew—especially for deep in-the-money calls—indicating elevated downside and systematic risk. In contrast, the more diversified Russell 2000 exhibits a more symmetrical smile, reflecting lower tail risk and a greater share of idiosyncratic components.

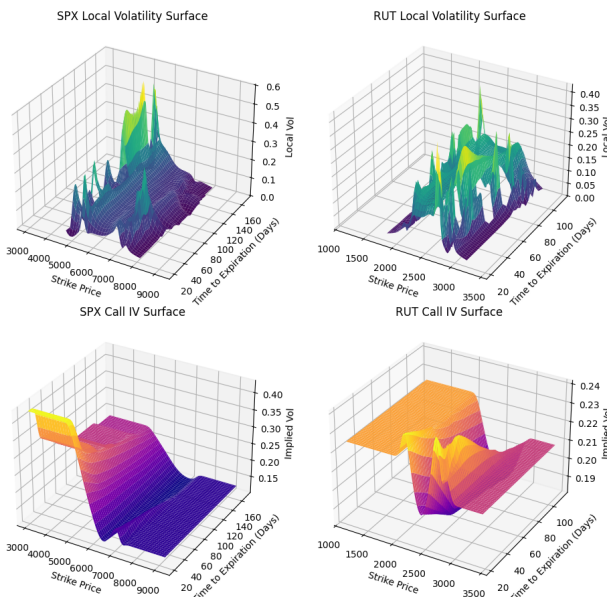


Figure 2: Comparing Local and Implied Volatility Surfaces for SPX and RUT

The SPX local volatility surface (top-left) dis-

plays sharp spikes in volatility at specific strike-maturity combinations. For options with maturities below 100 days, volatility is noticeably higher for deep in-the-money and out-of-the-money strikes. As the expiration period extends, these spikes shift deeper in-the-money and become more pronounced, reinforcing a stronger downside skew over longer horizons. This pattern aligns with the SPX implied volatility surface (bottom-left), which exhibits a steeper skew, particularly in far in-the-money strikes for near-term options. These findings suggest that index concentration effects heighten downside risk in the S&P 500 relative to more diversified indices like the Russell 2000.

Conversely, the Russell 2000’s local volatility surface displays more uniform spikes, while its implied volatility surface features a symmetrical near-the-money smile, reflecting lower tail risk. These observations confirm that index concentration in the SPX, driven by a few high-volatility stocks, leads to a steeper volatility skew compared to the diversified RUT, motivating a closer examination of index composition.

ii) Volatility and Weight Metrics for the SPX Basket Decomposition Table 3 summarizes the calibration of the Mag 7 basket over a three-month horizon. The complete volatility curve (Figure 3) is generated using the same method across all strikes.

Table 3: Summary of 3-Month ATM Implied Volatility and Weight Metrics for the Mag 7 and the S&P 500

Ticker	S_0	ATM Implied Vol (σ_{imp})	Normalized Weight	Relative Weight
AAPL	244.8	0.2476	0.214	0.0737
MSFT	415.57	0.2392	0.181	0.0622
AMZN	225.97	0.3249	0.131	0.0453
GOOG	187.0	0.3065	0.131	0.0451
META	700.2	0.3654	0.082	0.0283
NVDA	139.0	0.5334	0.184	0.0632
TSLA	360.83	0.5866	0.075	0.0259
Basket _{Mag7}	292.42	0.2678	1	0.3437
S&P500	6142.91	0.1405	–	–
Basket _{rest}	–	0.0900	–	0.6563

Table 3 summarizes key metrics for the Mag 7 constituents, including S_0 , ATM implied volatility, and both normalized and relative weights. For instance, AAPL exhibits an ATM volatility of approximately 24.76%, whereas NVDA and TSLA show significantly higher values (53.34% and 58.66%, respectively). The aggregated Mag 7 basket, obtained via a correlation-adjusted Monte Carlo simulation, yields an ATM volatility of 26.78%, compared to 14.05% for the full S&P 500

and 9.00% for the residual basket. Figure 3 illustrates that the Mag 7 basket (blue curve) displays pronounced downside risk at lower strikes, while the S&P 500 (orange) lies between the Mag 7 and the more stable residual basket (green), underscoring the elevated tail risk in concentrated, tech-heavy segments.

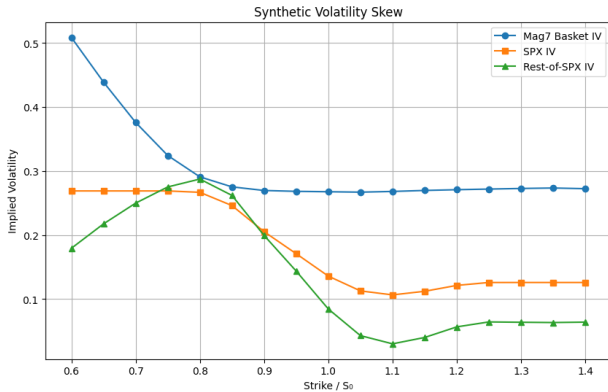


Figure 3: 3-Month Implied Volatility Curves for the Mag 7 Basket, S&P 500, and the Residual Basket

These findings underscore the critical role of index concentration and motivate our subsequent empirical analysis, which examines the dynamic behavior of the Mag 7 stocks and their impact on overall market volatility.

3.2 Empirical Analysis of Mag 7 Dynamics

The increasing dominance of the Mag 7 stocks has markedly reshaped S&P 500 market dynamics, particularly in the propagation of volatility as shown in previous frameworks. Their substantial market capitalization and outsized influence drive market behavior, especially during volatility spikes when their synchronized movements amplify broader stress. Historical data reveal key trends in how Mag 7-driven volatility interacts with SPX fluctuations, offering valuable insights for risk management, trading strategies, and market stability.

A. Mag 7 Volatility Spikes and the VIX

i) Methodology: We computed daily logarithmic returns to capture the return dynamics of the Mag 7 stocks and applied a 30-day rolling standard deviation—annualized to reflect yearly market fluctuations—to evaluate volatility. To capture longer-term trends, we also estimated a one-year rolling mean and standard deviation of the rolling volatility. Periods of significant market

turbulence were identified by calculating a z-score of the rolling volatility, with values exceeding two classified as volatility spikes. (Figure 18)

ii) Systemic Implications: The VIX index was plotted alongside periods of heightened Mag 7 volatility, with volatility spikes highlighted in red and weighted by the number of affected stocks. This visualization (Figure 4) illustrates how volatility spikes in the concentrated Mag 7 often align with surges in market-wide uncertainty, as measured by the VIX.

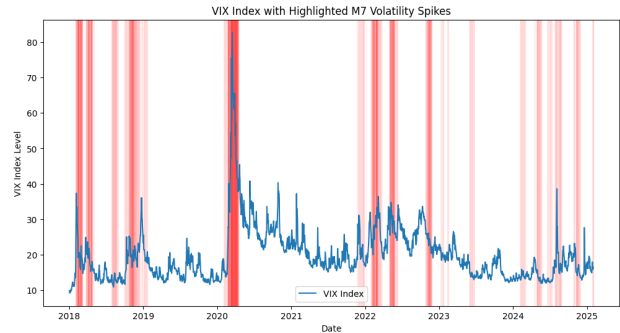


Figure 4: VIX During Mag 7 Volatility Spikes

iii) Findings and Relevance to Market Concentration: Our analysis shows that Mag 7 volatility spikes often align with VIX surges during major market stress, indicating a significant influence on overall market volatility. The varying intensity of these spikes suggests that the impact of the Mag 7 is most pronounced during broad market shocks, while persistent volatility in these stocks—even when the VIX is moderate—points to additional independent risk factors.

B. Inter-dependencies Within the Mag 7 During Volatility Events

i) Correlation Patterns: Under normal conditions, Mag 7 stocks exhibit moderate correlations—AMZN, MSFT, and META typically exceed 0.75, while TSLA and NVDA are less correlated, suggesting some independence. During volatility spikes, however, the correlation structure shifts markedly; MSFT, AAPL, and Alphabet display correlations exceeding 0.90, indicating strong co-movement under stress, whereas NVDA and META show lower correlations, implying more idiosyncratic behavior.

ii) VIX Relationships and Statistical Evidence: Correlations between Mag 7 stocks and the VIX

strengthen during market stress. Notably, MSFT, AAPL, and Alphabet reach correlations above 0.80 with the VIX during volatility spikes. Table 4 and Figure 19 presents t-test results that confirm significant differences in VIX levels during and outside these spikes.

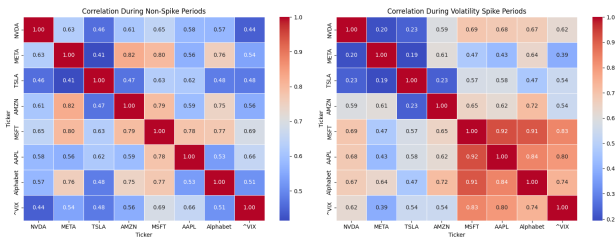


Figure 5: Mag 7 & VIX Correlations

Table 4: T-Test Results for VIX Levels

Metric	Value
Mean (During Spikes)	22.120699
Mean (No Spikes)	19.043228
T-Statistic	5.721722
P-Value	0.000000

iii) Implications for Market Risk: The increased correlation among key Mag 7 stocks during volatility spikes suggests a higher risk of synchronized declines, which may amplify systemic market stress. The heightened alignment between stocks like MSFT, AAPL, and Alphabet with the VIX during these periods calls for dynamic risk assessment, as diversification benefits observed under stable conditions may diminish during downturns, reinforcing the need for adaptive hedging strategies.

C. Mag 7 Correlation and S&P 500 Volatility
To examine the relationship between Mag 7 co-movement and market volatility, we computed a 60-day rolling correlation among Mag 7 stocks and compared it to SPX rolling volatility. The plot reveals that higher Mag 7 correlation aligns with increased SPX volatility, suggesting that synchronized movements within Mag 7 amplify broader market fluctuations. A significant positive correlation ($r = 0.7294$) between Mag 7 correlation and SPX volatility confirms this linkage. This underscores the systemic impact of Mag 7 stocks, where rising co-movement signals heightened market risk, reinforcing the need to monitor concentration dynamics for effective risk management.

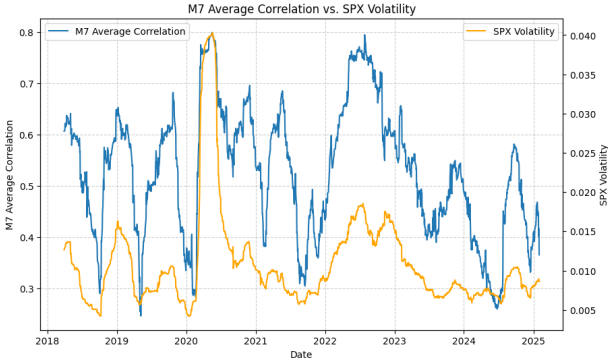


Figure 6: Mag 7 Correlation vs S&P 500 volatility

4 Implications for Long Small-Cap and Short Large-Cap Strategies

This section will explore the ramifications of index concentration for long small-cap/short large-cap strategies. In the traditional long small-cap/short large-cap strategy, the short leg typically uses a market-cap-weighted portfolio of large-cap stocks. When a few mega-caps dominate this portfolio, their idiosyncratic moves—especially in technology or AI-related events—can disproportionately drive the short leg’s performance. This creates two potential issues:

1. **Distorted Risk-Return Trade-off:** The presumed size premium may be distorted if the large-cap side is not a true diversified representation of broader market risk.
2. **Excess Volatility and Skew:** The short leg might be exposed to extreme moves due to the concentrated risk in a few names.

In a concentrated index, high-beta, high-volatility stocks tend to dominate the overall risk profile. We anticipate that the standard strategy may need to be adjusted to account for the elevated tail risk associated with the Mag 7. The traditional long small-cap, short large-cap strategy is based on the expectation that small-cap stocks will outperform large-cap stocks over time.

4.1 Proposed Strategy

Based on our findings that a significant portion of the S&P500’s systematic risk is concentrated in a small group of high-growth mega caps, we can isolate idiosyncratic returns from market beta. Our strategy combines long positions in these mega-cap equities and a small-cap index (IWM) with a short position in SPY to neutralize mar-

ket exposure and capture relative performance differences.

A. Methodology

We estimate beta using a rolling exponentially weighted mean of weekly log returns, which minimizes daily noise and captures medium-term sensitivity. We compute each asset's beta as:

$$\beta_i = \frac{\text{Cov}_w(r_i, r_{\text{SPY}})}{\text{Var}_w(r_{\text{SPY}})}.$$

The portfolio beta is the average beta of the long positions:

$$\bar{\beta}_{\text{long}} = \frac{1}{N} \sum_{i=1}^N \beta_i$$

Daily beta-neutral returns are calculated by subtracting the market component from the long portfolio's return:

$$r_{\text{portfolio}}(t) = \underbrace{\frac{1}{N} \sum_{i=1}^N r_{i,t}}_{\text{Long Portfolio Return}} - \bar{\beta}_{\text{long}} \times r_{\text{SPY},t}$$

We rebalance annually using the first available weekly date, ensuring that if the long portfolio has a beta of β_{net} and the short position in SPY has a beta of 1, the net beta is near zero.

B. Results

Over a 5-year backtest, the beta-neutral strategy achieved an annualized return of 14.98% with a Sharpe ratio of 0.61, though it experienced a 36% maximum drawdown, indicating that stock-specific risks remain. In contrast, the classical long small-cap/short large-cap strategy yielded a negative annualized return of approximately 7.6%, a Sharpe ratio of -0.38, and a deeper drawdown of -47.02%. These results highlight that including high-growth mega caps is crucial for isolating idiosyncratic returns and mitigating downside risk, outperforming the traditional approach both in absolute and risk-adjusted terms. This strategy can be extended to options trading by using options to hedge and enhance the beta-neutral portfolio. For example, traders could hold long positions in high-growth mega-caps and a small cap-index while shorting SPY, and then overlay options positions by buying puts on SPY or selling calls on the long equities to dynamically hedge against downside risk and capture volatility skew. With

dynamic rebalancing based on updated beta estimates and option-implied risk measures, this approach can more effectively isolate idiosyncratic returns while managing tail risk in evolving market conditions. Future research could explore optimal trading strategies for integrating such option overlays, including the use of machine learning techniques to forecast volatility changes and improve dynamic hedging accuracy.

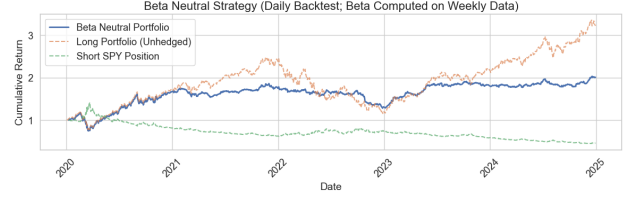


Figure 7: Modified Beta-Neutral Strategy

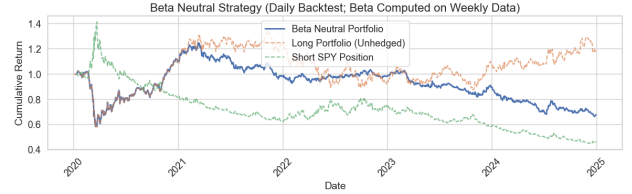


Figure 8: Traditional Long-Small/Short-Large Cap Strategy

5 Synthesis and Conclusion

This paper demonstrates that increasing market concentration—exemplified by the S&P 500's Magnificent Seven—profoundly affects systematic risk and option pricing. Our beta decomposition shows that the Mag 7 exhibit short-term betas rising from approximately 1.27 to 1.42–1.50 and contribute over 40% of the index beta, thereby eroding diversification. Our volatility analysis further indicates that the concentrated influence of these stocks steepens the volatility skew, with their basket displaying nearly double the ATM implied volatility of the broader index and strong correlations with market fluctuations. Moreover, our strategy backtests reveal that traditional long small-cap/short large-cap approaches are distorted by concentrated risk, while a beta-neutral strategy incorporating high-growth mega caps yields superior risk-adjusted returns. Collectively, these results suggest that conventional measures of “market risk” warrant reassessment and provide a basis for future research on global indices and alternative benchmarks.

5.1 Summary of Findings

Over a 5-year horizon, the Mag 7 portfolio exhibits a beta of approximately 1.27–1.28, rising to 1.42–1.50 in the short term, with their contribution to index beta increasing from roughly 36.5% to over 40%. This growing concentration diminishes diversification benefits, and technology stocks display higher return forecasts against the cap-weighted SPX relative to other sectors. Tesla's two distinct declines in 2024 further illustrate that even dominant stocks can experience pronounced idiosyncratic shocks. Our volatility analysis finds that the Mag 7 basket has an ATM implied volatility of 26.78%, significantly exceeding the S&P 500's 14.05% and the residual basket's 9.00%. Frequent alignment of Mag 7 volatility spikes with VIX surges—and a strong positive correlation ($r = 0.7294$) between Mag 7 co-movement and SPX volatility—confirms that their synchronized movements amplify overall market risk. Finally, our strategy analysis shows that traditional long small-cap/short large-cap strategies suffer from distorted risk-return trade-offs due to concentrated exposure. In contrast, our beta-neutral approach—combining long positions in high-growth mega caps and a small-cap index with a short SPY position—achieved an annualized return of 14.98% (Sharpe 0.61, max drawdown 36%) over 5 years, compared to -7.6% (Sharpe -0.38, drawdown 47.02%) for the traditional strategy. These findings emphasize the benefits of isolating idiosyncratic returns to mitigate downside risk in concentrated markets.

5.2 Implications for Investors

Our findings indicate that traditional measures of market risk may understate the influence of a few dominant stocks in index performance. Institutions should consider reassessing their portfolio exposures in light of the high contribution of the Magnificent Seven to overall systematic risk, particularly when using cap-weighted benchmarks. In practice, this may involve supplementing conventional risk models with analyses that explicitly quantify the impact of index concentration, as demonstrated in our beta decomposition and volatility studies. Such insights can inform more cautious asset allocation and hedging strategies, especially during periods of market stress. For retail investors, our analysis reveals

that many widely held index funds are effectively “top-heavy,” with a handful of mega-cap stocks driving much of the index's performance and risk. This concentration can lead to higher volatility and tail risk than typically assumed by a diversified market. Retail investors may benefit from a closer examination of index fund compositions and from considering strategies that balance exposure to mega-cap stocks with other market segments. An increased awareness of these dynamics can help investors make more informed decisions and tailor their portfolios to better manage potential downside risks.

5.3 Limitations and Future Research Directions

Our study has several limitations that warrant further investigation. First, the beta and volatility estimates, derived from historical data with exponential decay, may not capture abrupt structural shifts. Likewise, using historical correlations to approximate basket interdependence may oversimplify market behavior during extreme events. Additionally, the synthetic basket approach, grounded in risk-neutral models (Black–Scholes and Dupire), does not fully account for stochastic volatility or jump processes, potentially limiting pricing precision.

Future research should investigate dynamic basket-implied correlations, advanced option pricing models, global index applications, and machine learning techniques for enhanced volatility forecasting and risk management.

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A Appendix 1: Supplementary Figures

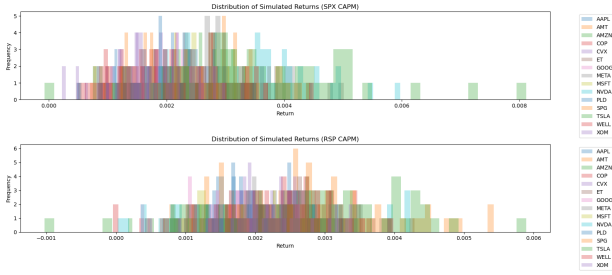


Figure 9: Return Distributions of GBM Stock Simulations using different CAPM Betas as drift.

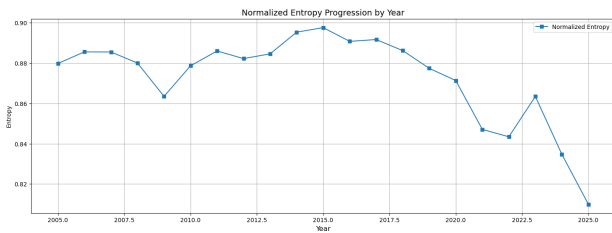


Figure 10: Normalized Entropy Over Time.

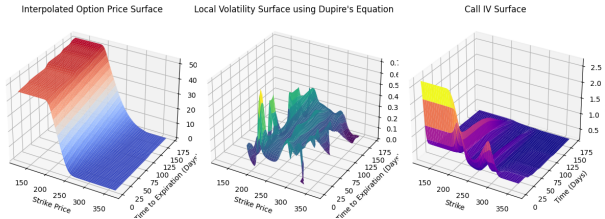


Figure 11: Apple Volatility Surfaces.

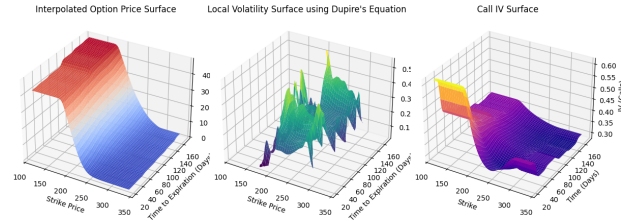


Figure 12: Amazon Volatility Surfaces.

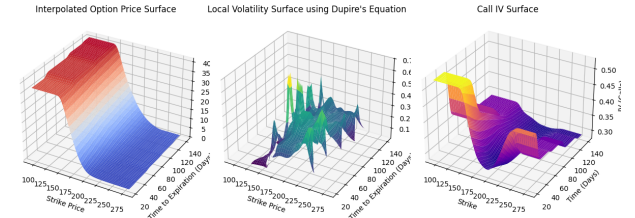


Figure 13: Google Volatility Surfaces.

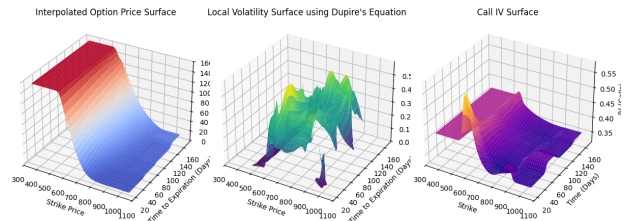


Figure 14: Meta Volatility Surfaces.

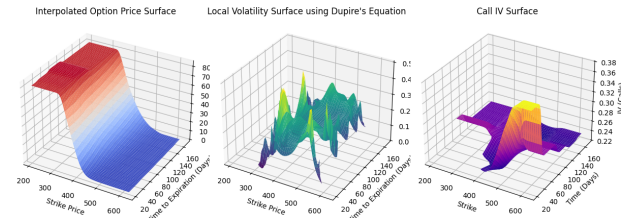


Figure 15: Microsoft Volatility Surfaces.

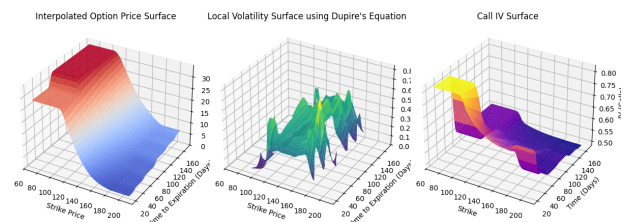


Figure 16: NVIDIA Volatility Surfaces.

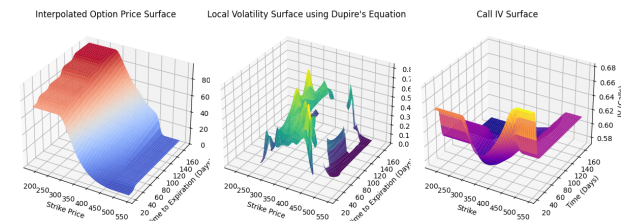


Figure 17: Tesla Volatility Surfaces.

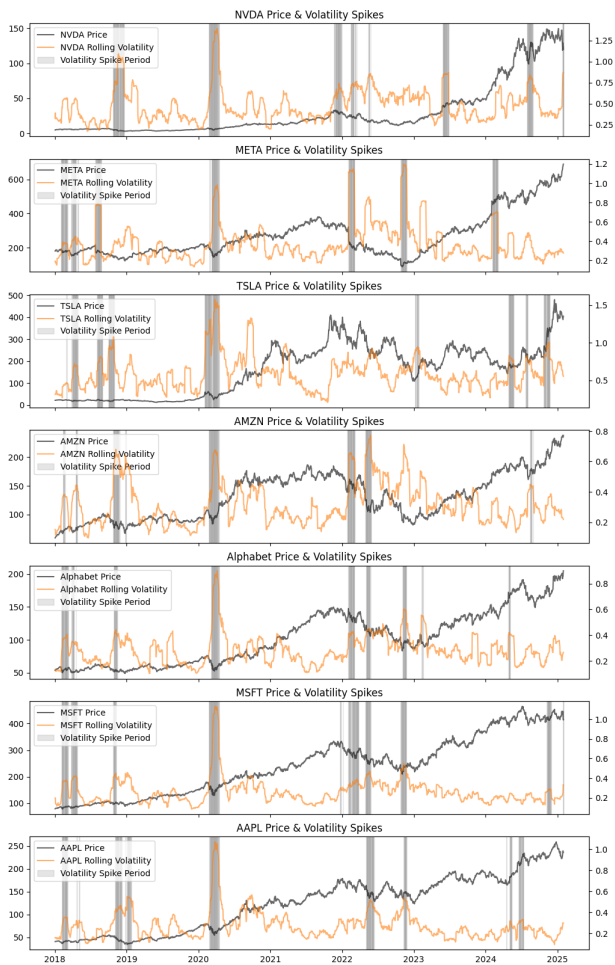


Figure 18: *Mag 7 Volatility Spikes.*

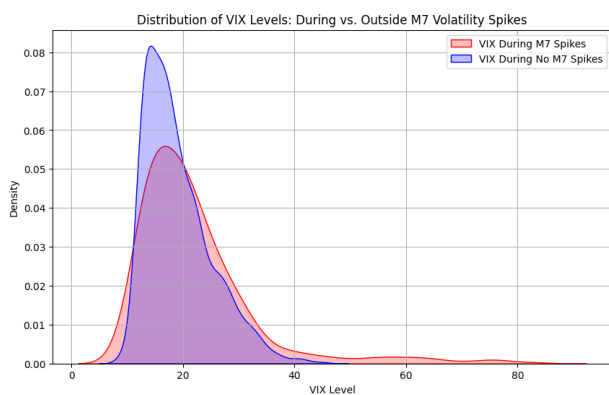


Figure 19: *VIX Level: During vs Outside Mag 7 Volatility Spikes.*